Tutorial 3

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Question 1

caseA: if $x=0$ or $y=0$, it is trival to be proved.

case B: Assume $\|x_p\|>0$, and $\|y_p\|>0$

Now we have: $\|X\|_p = \left[\sum_{i=1}^{n} |x_i|^p\right]^{\frac{1}{p}}$

considering "$\frac{1}{p}$" makes it difficult to analyse,

we can use triangle inequality

$\|x+y\|_p \le \|x\|_p + \|y_1\|_p$ to prove:

we let $t = \|x\|_p + \|y\|_p$, and $x_1 = \frac{x}{\|x\|_p}$, $\cancel{x_2}\ y_1 = \frac{\cancel{x}\ y}{\|y\|_p}$

∴ $\left\| \frac{\|x\|_p}{t} \cdot x_1 + \frac{\|y\|_p}{t} \cdot y_1 \right\|_p \le 1$

∴ ~~let~~ $\cancel{\lambda = \frac{\|x_p\|_p}{t},\ 1-\lambda}$

(~~bec~~

∴ Considering $1 = \frac{\|x\|_p}{t} + \frac{\|y\|_p}{t}$,

we let $\lambda = \frac{\|X\|_p}{t}$, $1-\lambda = \frac{\|y\|_p}{t}$

∴ $\|\lambda x_1 + (1-\lambda) y_1\|_p^p \le 1$

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Now we transfer them to functions, we have:

$$f(\lambda x_1 + (1-\lambda) y_1) \le \lambda f(x_1) + (1-\lambda) f(y_1)$$

where $f(x) = \|x\|_p^p$ is convex

Considering $f(x)$ is convex,

it is evident that $\|x\|_p = [\sum_{i=1}^{n} |x_i|^p]^{\frac{1}{p}}$

is a norm when $p \geq 1$

(I asked some students and schoolmates, is it related to Minkowski inequality or Hölder's inequality?)

Question 2

Now: $x \in \{x_1, \cdots\cdots, x_k\}$, $y \in \{x_1', \cdots x_k'\}$

∴ $p(x): (p_1 \cdots\cdots, p_k)$, $q_k: (q_1, \cdots\cdots q_k)$

Moreover ∵ $\sum p_i = 1$, $\sum q_i = 1$

∴ We need to prove:

∴ $DKL(p(x), q(x)) = \sum_{i=1}^{k} p_i \ln \frac{p_i}{q_i} \geq 0$

Let $f(x) = \ln(x)$ ∵ $f''(x) = -\frac{1}{x^2} < 0$

∴ $f(x)$ is concave

~~More~~ let $x_i = \frac{q_i}{p_i}$ ∴ $\ln \frac{p_i}{q_i} = -\ln x_i$

Moreover ~~$\sum p_i x_i = 1$~~ ∵ $\sum_{x \in X} p(x) = \sum_{x \in (X)} q(x) = 1$.

when $p(x) > 0$ and $q(x) > 0$ ∴ $\sum_{x \in X} p_i x_i = 1$

∴ $f(\sum p_i x_i) = f(1) = 0$

~~$\sum p_i$~~ Moreover ∵ $f(x)$ is concave, $x \in R$.

∴ $f(\sum p_i x_i) \geq \cancel{\sum p_i x_i} \, p_i f(x_i)$

Moreover ∵ $\sum p_i f(x_i) = \sum p_i \ln(x_i)$

$= -\sum p_i \ln \frac{p_i}{q_i}$

∴ $0 \geq -\sum p_i \ln \frac{p_i}{q_i}$

∴ $\sum p_i \ln \ln \frac{p_i}{q_i} \geq 0$

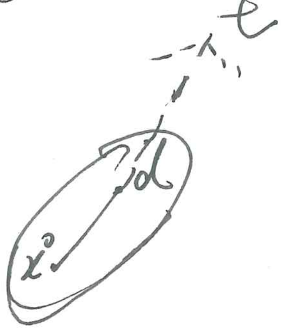
∴ $DKL(p(x), q(x)) \geq 0$

[In Information Theory, we have similar concept called "relative entropy", is it derived by Gibb's inequality as well?)

Directional Function:

$f(x)$ $\quad x \in R^n$

$x^0 \quad d \in R^n$



$g(t) = f(x^0 + td) \quad t \geq 0$

$$g'_+(0) = \lim_{t \to 0^+} \frac{g(t) - g(0)}{t} = \lim_{t \to 0^+} \frac{f(x^0 + td) - f(x^0)}{t}$$

$$= \frac{\partial f(x^0)^T}{\partial t} \cdot d$$

$$= \nabla f(x_0)^T \cdot d$$

$\nabla f(x^*)^T \cdot (y - x^*) \geq 0$

for any feasible direction at x^*

if $\nabla f(x_0)^T \cdot d = 0$ for any $d \in R^n$

$\Updownarrow$

$\nabla f(x_0^*)^T = 0$

补空间

$\nabla f(x^*)^T d > 0$ for all ~~fes~~ feasible d

$\Updownarrow$

$\nabla f(x^*)^T d \geq 0$ for any d with $Ad = 0$

$\Updownarrow$

$\nabla f(x^*)$ is a linear ~~comba~~ combination of $a_1, a_2, a_3 \cdots\cdots a_n$

$\Updownarrow$

$$\nabla f(x^*) = \underbrace{(a_1, a_2, a_3 \cdots\cdots a_n)}_{A^T} \underbrace{\begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}}_{-V}$$

$= A^T \cdot V$ $\quad x^* = (1, -\frac{1}{2}, 1)$

x^* is optimal for a convex ~~pattern~~ problem.

$\Leftrightarrow$ x^* is ~~a fesi~~ feasible and

$\nabla f(x^*)^T \cdot (y - x^*) \geq 0$

for any $y \in$ feasible region

$\nabla f(x^*) = Px^* + q = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

$\therefore \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}^T \begin{pmatrix} y_1 - 1 \\ y_2 - \frac{1}{2} \\ y_3 + 1 \end{pmatrix} > 0 \Rightarrow \therefore (1 - y_1) + 2(y_3 + 1) \geq 0$ for any feasible y